

Label Noise is the Bottleneck: A Multi-Stage Data Refinement Approach for Role-Playing Toxicity Detection

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Abstract

This paper addresses role-playing toxicity detection, classifying movie character dialogue as safe or unsafe. We argue that label noise in LLM-generated training data is the primary performance bottleneck. We propose a two-stage label refinement pipeline combining confident learning (Cleanlab) with LLM verification using Claude Sonnet. Our best system, a 5-fold DeBERTa-v3-base ensemble with optimized threshold of 0.57, achieves an out-of-fold F1 of 0.910 and Kaggle F1 of 0.828. Experiments show data quality improvements yield larger gains than multi-model ensembles or multi-seed training, and that aggressive LLM relabeling can paradoxically introduce new noise.

1 Introduction

Role-playing with AI characters has become prevalent, creating a need for systems that detect toxic dialogue in character interactions. Given a movie name and a role-playing scenario, a system must classify it as safe (label 0) or unsafe (label 1). The challenge is nuanced: the same language can be safe or unsafe depending on context. For example, a character criticizing an oppressive system (*"The system treats us like we are worthless"*) is safe, while directing the same sentiment at a person (*"You are worthless"*) is unsafe.

A key characteristic of this dataset is that labels were generated by LLMs rather than human annotators, introducing noise. We hypothesize that this label noise, not model architecture, is the primary performance bottleneck. To address this, we propose a two-stage label refinement pipeline combining statistical noise detection with LLM-based verification. Our experiments demonstrate that (1) targeted label cleaning provides larger gains than switching transformer architectures, (2) post-hoc threshold optimization (Jansche, 2005) on out-of-fold predictions provides consistent improvements, and (3) aggressive LLM relabeling can paradoxically degrade performance by introducing new disagreements with the evaluation distribution.

2 Related Work

2.1 Transformers for Text Classification

DeBERTa-v3 (He et al., 2023) introduces disentangled attention. RoBERTa (Liu et al., 2019) improves BERT through optimized pre-training. DistilBERT (Sanh et al., 2020) retains 97% of BERT's performance at 60% speed. We find DeBERTa-v3 most effective for this task.

2.2 Learning with Noisy Labels

Confident learning (Northcutt et al., 2021) identifies label errors by estimating the joint distribution between noisy and true labels. Gilardi et al. (2023) find LLMs can match crowd-worker annotation quality. However, LLM annotations can introduce systematic biases, motivating careful validation.

2.3 Threshold Optimization

Lipton et al. (2014) show that threshold optimization is mathematically necessary when maximizing F1. Jansche (2005) demonstrates threshold-moving yields consistent improvements under label noise. K-fold ensembles (Dietterich, 2000) provide robust out-of-fold estimates for threshold selection.

3 Methodology

3.1 Data Preprocessing

The dataset contains 6,410 samples across 90 movies (52.9% safe, 47.1% unsafe). We apply light normalization (smart quotes, whitespace, control characters) and remove 3 malformed rows, yielding 6,407 samples. Lowercasing is avoided to preserve signals transformers exploit. Movie and scenario are combined as *"Context | Movie: {name} Scenario: {text}"*. All samples fit within 512 tokens.

3.2 Label Noise Detection and Refinement

Our label refinement pipeline operates in two stages.

Stage 1: Confident Learning. We apply Cleanlab (Northcutt et al., 2021) using cross-validated probabilities from a 3-fold DeBERTa-v3-base classifier. Each sample receives a quality score (0 to 1). Samples below 0.3 have labels

flipped; the uncertain zone (0.3 to 0.5, 454 samples) is flagged for LLM review; samples above 0.5 are retained. This produces *train_confident*, the Cleanlab-refined dataset used by our best model.

Stage 2: LLM Verification. The 454 uncertain-zone samples are sent to LLMs for independent re-evaluation. We initially attempted verification using Groq's Llama-3.3-70B API, but encountered persistent rate limiting on the free tier (6,000 tokens per minute cap) that prevented batch completion. We subsequently used the Anthropic API with Claude Sonnet, which successfully processed all 454 samples in approximately 26 minutes. The prompt provides explicit definitions of safe and unsafe labels with carefully designed contrastive boundary examples distinguishing personal attacks from systemic criticism. Labels where Claude disagrees with the original are flipped, producing *train_confident_claude*. As we show in Section 4, this additional relabeling stage unexpectedly degraded downstream performance.

3.3 Model Architecture

We select DeBERTa-v3-base (He et al., 2023) as our primary model based on comparative experiments across three transformer architectures. DeBERTa-v3 employs disentangled attention, which separately encodes content and position using two independent vectors. In our experiments, DeBERTa-v3 achieves a validation F1 of 0.910, substantially outperforming RoBERTa at 0.893 and DistilBERT at 0.792. The disentangled attention mechanism is particularly effective for this task, where identical words carry different toxicity signals depending on their syntactic role (e.g., whether an insult targets a person directly or describes systemic mistreatment).

3.4 Training and Inference Strategy

We use stratified 5-fold cross-validation, training five DeBERTa-v3-base models on different 80/20 splits while maintaining label balance. Training uses AdamW (lr=1e-5, weight decay 0.01), linear warmup (10%), batch size 8 with gradient accumulation 2 (effective 16), max length 256, and 3 epochs per fold with checkpoint selection on validation F1.

At inference, fold predictions are averaged to produce final probability estimates. Following Lipton et al. (2014) and Jansche (2005), we perform post-hoc threshold optimization via grid search (0.30 to 0.70, step 0.01) on OOF

predictions. The optimal threshold 0.57 compensates for the model assigning borderline unsafe cases probabilities just below 0.5.

4 Experiments

4.1 Setup

All experiments are conducted on Google Colab with a Tesla T4 GPU (15.6 GB). The evaluation set contains 712 unlabeled samples scored via a Kaggle leaderboard. We report both OOF F1 (computed on training data via cross-validation) and Kaggle F1 (computed on the held-out evaluation set). We denote Cleanlab-only refined data as *train_confident* and data with additional Claude relabeling as *train_conf_claude*. Table 1 presents the full progression of our experiments across eight configurations.

Method	Data	OOF	Kaggle
TF-IDF+LogReg	conf	.801	.811
RoBERTa	conf	.891	.801
DeBERTa	conf	.910	.813
5-fold DeBERTa (t=.57)	conf	.910	.828
5f DeBERTa+ RoBERTa	conf claude	.909	.820
10f multi-seed DeBERTa	conf claude	.920	.821

Table 1: Results across all configurations. Bold indicates best Kaggle F1. *conf* = *train_confident* (Cleanlab only); *conf_claude* = *train_conf_claude* (+Claude relabeling). t = decision threshold.

4.2 Analysis

Label Noise is the Bottleneck. The most striking finding is that Claude Sonnet relabeling of uncertain samples degraded Kaggle performance rather than improving it. Models trained on *train_confident* (Cleanlab only) consistently outperform those trained on *train_conf_claude* (Cleanlab + Claude): the best Cleanlab-only model scores 0.828 versus 0.821 for the best Claude-refined model. This suggests that Claude's corrections, while individually plausible, shifted the training label distribution away from the evaluation set's ground truth. The result underscores that LLM relabeling is not universally beneficial and must be validated against downstream metrics rather than assumed to be an improvement.

Notably, the multi-seed 10-fold DeBERTa achieves the highest OOF F1 (0.920) across all experiments, yet its Kaggle F1 (0.821) is lower than the simpler 5-fold model (0.828). This confirms that higher internal cross-validation scores do not guarantee better generalization when the training labels themselves contain distributional differences from the evaluation labels.

Threshold Optimization. Consistent with Jansche (2005), shifting the decision threshold from 0.50 to 0.57 improves Kaggle F1 by 0.7 percentage points at zero additional training cost. The OOF F1-vs-threshold curve shows a smooth, unimodal peak at 0.57, confirming this is a robust optimum rather than an artifact of overfitting to a particular validation split. The upward shift indicates the model's predicted probabilities are slightly miscalibrated, assigning borderline unsafe cases probabilities just below 0.5.

Architecture vs. Data Quality. Among single models, DeBERTa-v3 (Kaggle F1 = 0.813) substantially outperforms RoBERTa (0.801) and the TF-IDF baseline (0.811). However, adding RoBERTa to the DeBERTa K-fold ensemble did not improve results (0.820 vs 0.828 for DeBERTa alone), as the weaker model's predictions diluted the stronger signal. This confirms that for this task, investing in data quality yields greater returns than broadening model diversity.

OOF vs. Kaggle Gap. A persistent gap of approximately 8 points exists between OOF F1 (~0.91) and Kaggle F1 (~0.83) across all models. Since K-fold cross-validation already controls for overfitting, this discrepancy likely reflects systematic differences in label noise patterns between training and evaluation sets, further supporting our central claim that data quality is the dominant challenge in this task.

5 Conclusion

We presented a multi-stage approach to role-playing toxicity detection that prioritizes data quality over model complexity. Our two-stage label refinement pipeline (confident learning followed by LLM verification) combined with 5-fold DeBERTa-v3 cross-validation and post-hoc threshold optimization achieves a Kaggle F1 of 0.828. Experiments across several configurations reveal that the simplest strong model on well-cleaned data outperforms more complex approaches on aggressively relabeled data. We also report the cautionary finding that LLM relabeling can degrade performance when the LLM's labeling criteria diverge from the evaluation distribution, highlighting the importance of validating every data intervention against downstream metrics.

Limitations

The OOF-Kaggle gap suggests unaddressed distribution shift. Our LLM verification uses a single model; multi-LLM voting may be more robust. Only base-sized models were evaluated due to GPU constraints. Confidence-based sample weighting during training, where uncertain samples receive reduced loss weights, could further improve robustness to label noise.

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